Supplementary Information — Junction gapmers across the *NR4A3* fusions of extraskeletal myxoid chondrosarcoma

Supplementary Information to *In silico, nearly half of junction-spanning 5-6-5 gapmer designs across 38 modelled NR4A3 fusion junctions of five extraskeletal myxoid chondrosarcoma partner genes pair a wild-type parent gene over a ten-base-pair duplex through the catalytic gap, and, as an identity between base counts, a longer gap trades gap-level margin against parent-paired gap DNA* — deposited as `fusion-junction-aso-research-article-manuscript.pdf`.

Preprint — not peer reviewed. Prepared for deposit; not yet posted. · Version of 2026-08-20 · supplementary information · built from 81749e7e7, tree not clean at build time

Running title. Junction gapmers across NR4A3 fusions — SI

Author. Tristan D. McRae

Independent researcher, unaffiliated. Correspondence: trimcrae@gmail.com

Research use only, and not for administration to any person or animal. Every oligonucleotide sequence named in this Supplementary Information, in the main text and in their tables is a research reagent intended solely for laboratory investigation. None is a medicine or a candidate drug, none has been synthesised or tested by anyone, and none may be administered to any human being or animal, compounded for such use, or supplied to any person for such use. Custom oligonucleotide synthesis is commercially available, so the restriction is on use rather than on access. The full statement is in the main text's Declarations.

Section numbers here are prefixed S. Cross-references of the form "§n" point to the main text (`fusion-junction-aso-research-article-manuscript.pdf`); "SI §Sn" points within this document. All numbered references are those of the main text's reference list, which is generated from the identifiers carried in the main text; the seven entries this document cites are repeated at its end so that an SI read on its own resolves them, and the full list is in the main text. This document carries no figure of its own: Supplementary Figure S1 is printed in the main text beside Figures 1 to 3 and carries its legend there, and it travels with the archive as well.

S1 · Target-site accessibility, and its exclusion from every ranking

§6 gives the estimator, the range across the panel and the fact that nothing in the main text is ranked on it. Here are the three reasons for that omission, in decreasing order of force.

First, the quantity is not the one the paper is about: accessibility predicts whether an oligonucleotide can reach its site on the fusion, and every question asked in the main text is whether it can be told apart from a parent once it does. An inaccessible site is a potency problem, and potency is not claimed for any sequence.

Second, the estimate is a fold of a naked transcript, whereas the compartment that matters is a nascent, protein-coated pre-mRNA; measured antisense activity correlates with such predictions weakly and inconsistently, which is why the field selects by screening rather than by folding.

Third, it would have no purchase on the result even if it were reliable: the surviving candidates are separated by orders of magnitude of predicted off-target load, and reordering them on a quantity that spans 0.160 to 0.707 across the panel would substitute a weak predictor for a strong one. The values are released so that a laboratory ordering several oligonucleotides at one junction can break a tie on them, which is the use they support.

S2 · The melting-temperature cross-check on the duplex thermodynamics

The 250 nM strand concentration used in the main text's thermodynamic scoring enters only the melting temperature, which is the quantity the independent implementation of §6 was compared against. The strand choice that check does not verify is fixed instead by the documented convention of the software interface the parameters are read through — Biopython's `R_DNA_NN1` table in `Bio.SeqUtils.MeltingTemp`, whose calling convention is that the sequence supplied is the RNA one, which is the sequence the scoring code supplies. The nearest-neighbour parameters themselves are Sugimoto and colleagues',[53] read from that table rather than entered by hand, and the package version they were read at is recorded beside every released free energy. The parameter set and the strand convention have different sources and are stated separately here for that reason.

S3 · Provenance of the three gap-length figures

§6 states the three figures that place a six-nucleotide gap below the reported optimum, and records that none of them is a titration in this architecture. Their provenance is as follows.

- **A minimum of five.**[41] Given in passing, citing prior work rather than measuring it.
- **A minimum of six.**[42] A design-protocol statement, likewise given in passing and citing prior work.
- **An optimum of seven to ten.**[39] Both this and the six-nucleotide activity figure beside it are taken from a review that credits them to named earlier primary studies rather than measuring them itself.

S4 · The graded re-score: which screens carry it

The re-score of §6 covers all 38 junction screens, and 39 of the 93 screens released in total. Two classes of screen are released ungraded. One coverage-only control screen records no gap-mismatch depth and so cannot be graded at all; the 53 deeper re-screens are released ungraded because the graded model adds nothing where no hit list is truncated. The re-score counts only sense-strand hits where it can, meaning where the retained hit list is complete and every hit's strand is therefore known.

Which of the two bounds applies to a given design follows from its hit list, and each design records that. Screens produced before the orientation fix carry a strand-blind count for every truncated design, because the strand of an unretained hit is unrecoverable; screens produced afterwards carry an already-filtered one.

S5 · The two unfiltered control screens

The two screens §6 records as unfiltered are modelled control junctions built in amino-acid rather than transcript coordinates. They carry no junction from the 38-junction panel, and no count reported anywhere in the main text is taken from them.

S6 · The coverage ladder's second basis, and four zero-contribution junctions

Table 5 prices every rung on the breakpoint distribution of a single 18-case series,[22] which is the basis 68.4% is computed on. Every figure in this section is arithmetic over published cohorts rather than a coverage measurement — no patient was screened with any sequence named here — and §4.1 states in full what that figure is, what it is not, and the four bounds on it. Two things sit around that ladder: where its arithmetic runs out, and a second basis that answers a different question and supersedes nothing.

The arithmetic runs out before the target does. A set covering every *EWSR1* and every *TAF15* breakpoint reaches 94.8% of molecularly confirmed cases and stops, so no panel restricted to those two partners reaches 95%, and the remaining reachable cases are the two *TCF12* tumours that the fourth reagent of §4.1 addresses.

Nine junctions now carry a published exon-resolved breakpoint and a screened design, eight of them through all five screens and the ninth, the *PGR* seam, through three of the five: its pre-mRNA compartment is unmeasured for the reason main text §2.6 gives, and so is its mature-parent compartment, because that screen reads the same committed six-gene sequence cache the pre-mRNA screen reads and that cache carries no *PGR*. One absent donor therefore costs two screens rather than one. Four of the nine sit outside the 38-junction panel, at the *NR4A3* exon-2 acceptors of §2.6. Four of the nine move the estimate; one moves only the bound, its arm having no measured within-partner distribution; and four move it by nothing at all — two because their partner is absent from this cohort, and two because their partner is present while their exon pair carries no count in it. That last pair is the easier of the two kinds of zero to miss, and a membership test that asked only about the partner did miss it. Priced on a pooled breakpoint basis rather than on the single series the ladder uses, the nine together are 82.9% of molecularly confirmed cases, widening to 57.5–90.7%. That range is built the way §4.1 builds its own and carries the same caveat: each breakpoint fraction is taken to its own Wilson bound while the partner shares are held at their point estimates, so it is a composed-endpoint range carrying no nominal coverage level, and it is not a confidence interval. The two reagents of §4.1 are 67.1% rather than the ladder's 68.4%. The denominators are not the ladder's, and neither are the numerators: the *EWSR1* arm is 17 of 20 here against the ladder's 10 of 15, and two changes make that move rather than one. The denominator widens by pooling the 18-case series with the five-case whole-transcriptome cohort,[25] and the numerator rises because this panel carries four *EWSR1* junctions where the ladder carries one, reaching 12 of the same 15 tumours before any pooling. The *TAF15* arm stays at 3 of 3. Three things follow. Fifteen of those 20 tumours come from one series, so the pool stays close to it; the two series agree at *EWSR1* exons 12 and 13 and disagree completely at exon 7 to *NR4A3* exon 2, 0 of 15 against 1 of 5; and a third series was refused because every case in its *EWSR1* arm carries a covered junction by construction, 12 of 12, since a case with any other junction could not have entered that arm at all; the series that is pooled is 12 of 15 on the same test and so is not fixed by its own assay. That third series is Okamoto and colleagues' 18 cases, PMID 11679947, cited only here and so taking no number in the main text's list. Pooling it would give 87.4% rather than 82.9%, which is to say the rule costs coverage rather than buying it, and it is applied for that reason and not for the figure. Two of seven retrieved series resolve every case to an exon pair, so this is a pooled record and not the whole one; the seven are the papers that report an exon-resolved junction at all in the breakpoint census behind this section, which read 295 papers and is the one home for both counts. The figure supersedes nothing: it answers a different question from the ladder, whose rungs are incremental and priced on the single series 68.4% is computed on, and 68.4% remains the coverage of the two reagents named in §4.1. Its own membership rule is an evidence test rather than a list — a junction qualifies where a published report places a patient's breakpoint at it and a reagent has been through all five screens, three of the five for the *PGR* seam, whose pre-mRNA and mature-parent compartments are both unmeasured rather than clean, on the one shared cache main text §2.6 describes — and one qualifying junction, *PGR* exon 2 to *NR4A3* exon 2, reported in a single patient,[28] moves the figure by exactly zero, because the 58-case cohort behind the denominator contains no *PGR* case for such a reagent to engage. The further caveat on that reagent — a sixth partner, outside the five the panel models, whose seam is screened against the non-canonical-acceptor table rather than through the panel's transcript models — is in §2.6, which also points at the accession §6 lists for it. *TFG* exon 7 to *NR4A3* exon 3 is the second of that pair, on the same ground and from a deposited cDNA rather than a report. Those two are the partner-absent half of the four that add nothing; the other two have their partner present in the cohort and their exon pair carrying no count in it. What such a reagent changes is which patients are reachable at all, which is a different statement and is not added to a coverage percentage.

References cited in this Supplement

These are the seven entries of the main text's reference list that this document cites, repeated here verbatim from it so that a reader holding only the Supplementary Information can resolve them. The numbers are the main text's and are assigned there; each entry carries its PubMed identifier in the same non-rendering comment the superscripts use, so the renumbering that keeps a superscript and its reference together reaches this block too. This is a copy, not a second list: the reference list of record is the main text's, generated from the identifiers carried in it.

[22] Panagopoulos I, Mertens F, Isaksson M, Domanski HA, Brosjö O, Heim S, Bjerkehagen B, Sciot R, Dal Cin P, Fletcher JA, Fletcher CD, Mandahl N. Molecular genetic characterization of the EWS/CHN and RBP56/CHN fusion genes in extraskeletal myxoid chondrosarcoma. Genes, chromosomes & cancer. 2002;35(4):340-352. PMID: 12378528. doi:10.1002/gcc.10127

[25] Urbini M, Indio V, Astolfi A, Tarantino G, Renne SL, Pilotti S, Dei Tos AP, Maestro R, Collini P, Nannini M, Saponara M, Murrone L, Dagrada GP, Colombo C, Gronchi A, Pession A, Casali PG, Stacchiotti S, Pantaleo MA. Identification of an Actionable Mutation of KIT in a Case of Extraskeletal Myxoid Chondrosarcoma. International journal of molecular sciences. 2018;19(7):E1855. PMID: 29937513. doi:10.3390/ijms19071855

[28] Wilbur HC, Robinson DR, Wu YM, Kumar-Sinha C, Chinnaiyan AM, Chugh R. Identification of Novel PGR-NR4A3 Fusion in Extraskeletal Myxoid Chondrosarcoma and Resultant Patient Benefit From Tamoxifen Therapy. JCO precision oncology. 2022;6:e2200039. PMID: 36103645. doi:10.1200/po.22.00039

[39] Kauppinen S, Vester B, Wengel J. Locked nucleic acid (LNA): High affinity targeting of RNA for diagnostics and therapeutics. Drug discovery today. Technologies. 2005;2(3):287-290. PMID: 24981949. doi:10.1016/j.ddtec.2005.08.012

[41] Mejzini R, Caruthers MH, Schafer B, Kostov O, Sudheendran K, Ciba M, Danielsen M, Wilton S, Akkari PA, Flynn LL. Allele-Selective Thiomorpholino Antisense Oligonucleotides as a Therapeutic Approach for Fused-in-Sarcoma Amyotrophic Lateral Sclerosis. International journal of molecular sciences. 2024;25(15):8495. PMID: 39126066. doi:10.3390/ijms25158495

[42] Agrawal S. Transient Cyclic Structured Oligonucleotide Designs for Therapeutic Applications. Current protocols. 2026;6(2):e70319. PMID: 41614678. doi:10.1002/cpz1.70319

[53] Sugimoto N, Nakano S, Katoh M, Matsumura A, Nakamuta H, Ohmichi T, Yoneyama M, Sasaki M. Thermodynamic parameters to predict stability of RNA/DNA hybrid duplexes. Biochemistry. 1995;34(35):11211-11216. PMID: 7545436. doi:10.1021/bi00035a029

One further source is cited only here and therefore takes no number in that list, exactly as the external data records do: Okamoto S, Hisaoka M, Ishida T, Imamura T, Kanda H, Shimajiri S, Hashimoto H. Extraskeletal myxoid chondrosarcoma: a clinicopathologic, immunohistochemical, and molecular analysis of 18 cases. Human pathology. 2001;32(10):1116-1124. PMID: 11679947. doi:10.1053/hupa.2001.28226 — the third breakpoint series SI §S6 records as refused.